

Freight Rate Prediction Assessment Report

1. Objective

The task is to train on `train-test.csv`, predict all 12,000 loads in `validation.csv`, complete the provided December scenario, and preserve the required output schemas.

2. Data exploration and quality

The labeled development data contains 48,000 loads from 1 January through 31 October 2025. The target is `posted_rate`. The main predictive signal is distance: its Pearson correlation with the target is approximately **0.91**. Equipment, route geography, market conditions, quote signal, date, and weight add useful context.

The data contains 300 missing weights (0.63%) and 374 missing market-index values (0.78%). There are also non-positive weights, including negative values, which are physically invalid for freight loads. The inference set has the same type of missingness. The target includes a small number of very large rates, so the final model uses MAE loss rather than allowing those few observations to dominate a squared-error objective.

3. Validation and split

Because the submission set represents future dates, I used a chronological holdout rather than a random split:

- **Training:** 1 January–31 August 2025
- **Validation:** 1 September–31 October 2025

This evaluates the ability to generalize forward in time and avoids using future market observations to predict earlier loads. On this holdout, the selected configuration achieved approximately **MAE 121.9** and **RMSE 637.1** in the local experiment. RMSE is higher because of a small number of extreme target values; the median absolute error was approximately 43.5.

4. Feature engineering and model

The model is a `CatBoostRegressor` with MAE loss, depth 7, learning rate 0.04, 360 trees, and L2 regularization. CatBoost was chosen because it can directly learn from categorical variables and nonlinear interactions while handling missing numeric values.

Features include:

- Pickup, delivery, equipment, and combined pickup-delivery route.
- Pickup and delivery latitude/longitude.
- Distance and `log1p(distance)`.
- Weight, market index, and quote signal.
- Month, day of week, day of month, day of year, ISO week.
- Cyclic annual sine/cosine date features.
- Missingness and invalid-weight indicators.

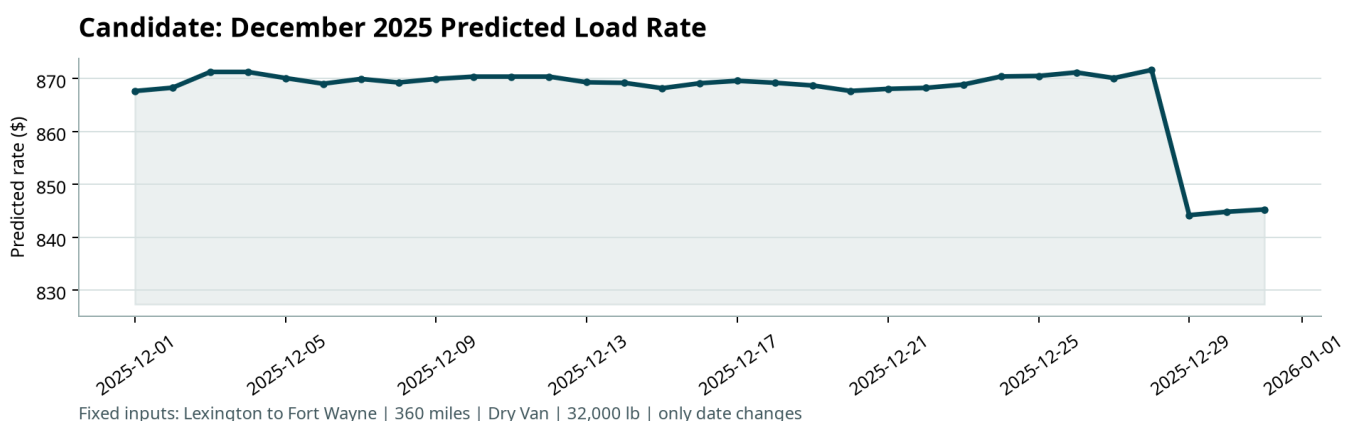
For December rows, columns that are not present in the supplied scenario file are safely aligned to the training feature schema as missing values. Predictions are clipped to a small positive floor because the scorer requires strictly positive rates.

5. Output validation

The provided scorer was run successfully. It validated:

- 12,000 unique validation IDs from `TE-000001` through `TE-012000`.
- Exactly two prediction columns in the required order: `load_id, predicted_rate`.
- 31 December rows covering every date from 1–31 December 2025.
- Positive rates and unchanged fixed December inputs.

The scorer generated the fixed chart below.



6. Reproducibility

Install dependencies and run:

```
python -m pip install -r requirements.txt
python solution.py
python score.py --predictions validation_predictions.csv --december-prediction
```